

FOREST CHANGE INTELLIGENCE FOR INDIA

VanDrishti

From coordinates and dates to field-ready forest evidence.

VanDrishti automatically retrieves satellite observations, detects likely forest loss and gain, and converts the result into maps, metrics, hotspots and web-ready geospatial outputs.

The platform removes the need for users to find or align satellite files. A forest officer, NGO or analyst can draw an area on the India map, select two periods and receive a decision brief.

10 m

SENTINEL-2 DETAIL

3

CHANGE CLASSES

2

SATELLITE CATALOGUES

EXECUTIVE SUMMARY

What the project does

1. Select

Choose a forest preset or draw a red analysis box anywhere over India.

2. Compare

Select before and after periods. The system finds suitable low-cloud scenes.

3. Detect

Cloud-masked NDVI change logic identifies forest loss, gain and stable areas.

4. Act

Review field priority, hotspots, confidence, carbon exposure and exportable evidence.

Primary users: forest departments, conservation NGOs, environmental researchers, restoration teams, district administrators and web platforms that need a forest-change API.

SYSTEM WORKFLOW

What VanDrishti uses and how it works



STAGE	WHAT HAPPENS
1. Area of interest	A user draws a rectangle or selects a preset Indian forest landscape. The area is limited to 2,500 km ² for responsive processing.
2. Date windows	Before and after ranges are selected. Similar seasons are recommended to reduce monsoon and phenology effects.
3. Satellite discovery	The API searches Sentinel-2 Level-2A scenes by bounds, date, cloud cover and coverage. It retries and uses a second catalogue if required.
4. Spectral processing	Blue, green, red, near-infrared and Scene Classification Layer data are read. Clouds, shadows, snow and invalid pixels are removed.
5. Change logic	NDVI forest classes are compared. A meaningful NDVI shift and minimum patch size are required, reducing small noisy detections.
6. Geographic evidence	Pixels are converted into area statistics, transparent overlays and GeoJSON polygons positioned on the India map.

Data and analytical components

Sentinel-2 Level-2A

Surface-reflectance imagery at 10 m for the visible and near-infrared bands used by the project.

NDVI

A vegetation signal calculated as $(\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$. Dense healthy vegetation generally has higher values.

Cloud classification

Sentinel-2 SCL pixels exclude cloud, cirrus, shadow, snow, saturated and no-data regions.

Hybrid model path

The live baseline uses explainable NDVI/RGB rules. A six-channel U-Net is included for training on labelled pairs.

OUTPUTS

What the system produces

OUTPUT	DESCRIPTION
Forest loss	Area in km2 and hectares, red polygons and a red transparent map overlay.
Forest gain	Area in km2, green polygons and a green transparent map overlay.
Net and annualised change	Gain minus loss, percentage changed and an annualised loss estimate based on observation dates.
Forest-cover trend	Estimated forest-like cover before and after, plus mean NDVI movement.
Loss hotspots	Up to five largest loss polygons ranked with coordinates and area for field verification.
Data quality	Valid-pixel percentage, scene cloud cover, observation gap, seasonal gap and comparison quality.
Action insight	Low, medium or high field priority with a recommended next action.
Carbon exposure	Indicative low-to-high tonnes CO2e range based on detected loss. It is explicitly presented as a screening estimate.
Exports	GeoJSON change polygons, CSV summary, printable report and JSON API response.

Map layers available to the user

Change detection

Red loss and green gain mask with adjustable opacity.

NDVI intensity

Continuous red-to-green strength of vegetation decline or recovery.

Before observation

Selected source scene preview aligned to the area bounds.

After observation

Later source preview for direct visual comparison.

Example API result

```
summary.forest_loss_sq_km 0.0883
summary.forest_loss_hectares 8.83
insights.field_priority medium
insights.comparison_quality strong
source.before.date 2023-02-17
source.after.date 2025-02-21
```

Values above illustrate the response structure. Actual values depend on the selected area, dates, cloud conditions and sensitivity settings.

DIFFERENTIATION

What makes VanDrishti unique

UNIQUE CAPABILITY	WHY IT MATTERS
No satellite-file handling	The main workflow needs only an area and dates. Scene discovery, access and processing happen automatically.
Dual-provider resilience	Planetary Computer is the primary catalogue; Element 84 Earth Search provides automatic fallback when required.
Detection plus decision support	The result is not just a mask. It includes confidence, field priority, hotspots, source quality and recommended action.
India-first interface	The map, forest presets, area controls and default experience are designed around Indian monitoring use cases.
Explainable baseline	NDVI thresholds, meaningful-change requirements, cloud masking and minimum patch controls are understandable to judges and users.
Interoperable outputs	GeoJSON, CSV, PNG overlays and an API allow integration with GIS, mobile field apps and government dashboards.
Fast repeat analysis	Before and after scenes load in parallel. A 32-observation cache makes repeated comparisons nearly immediate.

Practical applications

USER	PRACTICAL VALUE
Forest departments	Screen large areas, prioritise inspections and attach polygons to field assignments.
Conservation NGOs	Track restoration, encroachment and habitat pressure without operating a full remote-sensing stack.
District administration	Compare suspected clearing with permits, fires, roads and local land-use records.
Researchers	Export repeatable geospatial results and scene metadata for further validation.
Web and mobile teams	Call a documented REST API and display map-ready outputs without implementing satellite processing.

PRACTICAL SCENARIO

A forest officer draws a red box over a suspected clearing, compares January 2023 with January 2025, and receives ranked red hotspots. The largest polygon can be exported as GeoJSON and sent to a field team with its coordinates and estimated area.

IMPLEMENTATION

Technology stack and API

LAYER	TECHNOLOGY	ROLE
Frontend	HTML5, CSS3, JavaScript	Responsive dashboard, controls, reports and exports.
Interactive mapping	Leaflet, Leaflet Draw, OpenStreetMap	India map, AOI drawing/editing, layers, hotspots and overlays.
Backend API	Python, FastAPI, Uvicorn	Validation, endpoints, orchestration, CORS and OpenAPI documentation.
Satellite discovery	PySTAC Client	Spatial, temporal, cloud-cover and collection searches.
Satellite sources	Planetary Computer, Earth Search	Primary and fallback access to Sentinel-2 Level-2A.
Raster processing	Rasterio, NumPy, Pillow	COG windows, spectral calculations, masks, previews and overlays.
Computer vision	OpenCV	Morphology, connected components, contours and polygon simplification.
ML path	PyTorch U-Net	Optional six-channel before/after semantic segmentation model.
Testing	Pytest, FastAPI TestClient	API, validation, NDVI noise and geometry checks.
Packaging	Docker	Portable local or cloud deployment.

Main API routes

ROUTE	PURPOSE
POST /api/v1/detect/satellite	Automatic area-and-date Sentinel-2 analysis.
POST /api/v1/detect	Advanced before/after image upload analysis.
GET /api/v1/health	Service health, active engine and cache status.
GET /api/v1/model	Model classes, input requirements and available features.
GET /docs	Interactive OpenAPI documentation for web developers.

Responsible-use limitations

- Optical imagery is affected by residual clouds, haze, shadows, flooding and seasonal vegetation cycles.
- Plantations, crops and natural phenology can appear as forest gain or loss; similar-season comparisons improve reliability.
- Area and carbon figures are screening estimates. Regulatory decisions require local records and field validation.
- The current NDVI baseline should be benchmarked against labelled Indian forest-change data before operational deployment.